

In data analysis, we often start by examining a single column at a time. However, real-world scenarios frequently require us to analyze multiple columns simultaneously to extract meaningful insights. For example, determining how many apps in the *Art & Design* category have a rating of 4.5 or more involves considering both the *Category* and *Rating* columns.

This article will explore how to perform multi-column data analysis using Python and pandas. We will analyze a dataset of Google Play Store apps to answer key questions, such as:

- How many free apps exist in a particular category?
- How many apps have a high rating?
- How can we filter apps based on multiple criteria?

Preparing the Dataset

Before we begin the analysis, we must **prepare the dataset** by handling missing values.

```
import numpy as np
```

```
import pandas as pd
```

```
from sklearn.impute import SimpleImputer
```

```
# Load the dataset
```

```
df = pd.read_csv('googleplaystore.csv')
```

```
# Handle missing values in the Rating column by replacing NaNs with the mean rating
```

```
imputer = SimpleImputer(missing_values=np.nan, strategy='mean')
```

```
df.iloc[:, 2:3] = imputer.fit_transform(df.iloc[:, 2:3])
```

```
# Drop remaining NaN values
```

```
df = df.dropna()
```

Now, the dataset is **clean** and ready for analysis.

Understanding the Dataset Structure

Before we start filtering data, let's understand how the dataset is structured.

```
df.head()
```

Dataset Columns

- App: Name of the application.
- Category: The category of the app (e.g., ART_AND_DESIGN, FAMILY, GAME, etc.).
- Rating: The user rating.

- Reviews: Number of reviews received.
- Size: App size.
- Installs: Number of installs.
- Type: Free or Paid.
- Price: Cost of the app.
- Content Rating: Age group suitability.
- Genres: App genres.
- Last Updated: Last update date.
- Current Ver: Current version of the app.
- Android Ver: Minimum required Android version.

Now that we understand our data, let's start analyzing multiple columns.

Output:

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19M	10,000+	Free	0	Everyone	Art & Design	January 7, 2018	1.0.0	4.0.3 and up
1	Coloring book moana	ART_AND_DESIGN	3.9	967	14M	500,000+	Free	0	Everyone	Art & Design;Pretend Play	January 15, 2018	2.0.0	4.0.3 and up
2	U Launcher Lite – FREE Live Cool Themes, Hide ...	ART_AND_DESIGN	4.7	87510	8.7M	5,000,000+	Free	0	Everyone	Art & Design	August 1, 2018	1.2.4	4.0.3 and up
3	Sketch - Draw & Paint	ART_AND_DESIGN	4.5	215644	25M	50,000,000+	Free	0	Teen	Art & Design	June 8, 2018	Varies with device	4.2 and up
4	Pixel Draw - Number Art Coloring Book	ART_AND_DESIGN	4.3	967	2.8M	100,000+	Free	0	Everyone	Art & Design;Creativity	June 20, 2018	1.1	4.4 and up

So let us get started with the analysis now. Let us try to think about some questions which will be beneficial. So we can easily solve these problems if our fundamental knowledge related to pandas library is strong.

Q1. How many free apps are there in ART_AND_DESIGN?

To count the number of free apps in the Art & Design category, we need to filter the dataset based on the Category and Type columns.

```
df = df.values # Convert DataFrame to NumPy array
```

```
count = 0
```

```
for i in df:
```

```
    if i[1] == 'ART_AND_DESIGN' and i[6] == 'Free':
```

```
count += 1
```

```
print("There are", count, "free apps from ART_AND_DESIGN")
```

Output:

There are 61 free apps from ART_AND_DESIGN

So we can get the inference from here that in our dataset we don't have any apps which are free and belonging from the ART_AND_DESIGN category. So like this we can have many more questions as well.

Q2. How many apps are there in ART_AND_DESIGN with rating more then 4.5?

This time, we analyze both category and rating.

```
c = 0
```

```
for i in df:
```

```
    if(i[1] == 'ART_AND_DESIGN' and i[2] > 4.5):
```

```
        c += 1
```

```
print("There are",c,'apps from ART_AND_DESIGN with rating more than 4.5')
```

Output:

There are 22 apps from ART_AND_DESIGN with rating more than 4.5

Q3. How many apps are there in FAMILY with rating more then 4.5 and Free?

Now, we introduce a third condition: the app must be free.

```
c = 0
```

```
for i in df:
```

```
    if(i[1] == 'FAMILY' and i[2] > 4.5 and i[6] == 'Free'):
```

```
        c += 1
```

```
print("There are",c,'free apps from ART_AND_DESIGN with rating more than 4.5')
```

Output:

There are 314 free apps from ART_AND_DESIGN with rating more than 4.5

Q4. List all the free apps with rating more then 4.5 and category is FAMILY?

Instead of just counting, let's list all app names that meet our conditions.

```
print("*" * 100)

print("    These are the free apps where Category is FAMILY with rating more than 4.5.")

print("*" * 100)

for i in df:

    if i[1] == 'FAMILY' and i[2] > 4.5 and i[6] == 'Free':

        print(i[0])

print("-" * 100)
```

Output:

```
*****
*****

    These are the free apps where Category is FAMILY with rating more than 4.5.
*****
*****

U Launcher Lite – FREE Live Cool Themes, Hide Apps
Kids Paint Free - Drawing Fun
Mandala Coloring Book
Photo Designer - Write your name with shapes
ibis Paint X
Superheroes Wallpapers | 4K Backgrounds
HD Mickey Minnie Wallpapers
Harley Quinn wallpapers HD
Colorfit - Drawing & Coloring
I Creative Idea
UNICORN - Color By Number & Pixel Art Coloring
PIP Camera - PIP Collage Maker
Canva: Poster, banner, card maker & graphic design
Install images with music to make video without Net - 2018
Cardi B Wallpaper
```

X Launcher: With OS11 Style Theme & Control Center
AJ Styles HD Wallpapers
Fantasy theme dark bw black building
Spring flowers theme couleurs d t space

Data Analysis using Conditions

Previously, we discussed how to analyze multiple columns using **for loops** and **indexing** to filter data and find the required values. We iterated over each row, checked conditions manually, and counted or displayed relevant records.

However, **pandas** provides built-in functions that allow us to perform the same operations with just **one line of code** using **conditions**. These functions make data analysis much simpler and more efficient.

In this article, we will explore how to:

- Use **conditions in pandas** for data filtering.
- Answer key questions using conditions.
- Compare this approach to the previous method of **loop-based filtering**.

Dataset Preparation

Before applying conditions, let's first **load and clean the dataset**.

```
import numpy as np
```

```
import pandas as pd
```

```
from sklearn.impute import SimpleImputer
```

```
# Load the dataset
```

```
df = pd.read_csv('googleplaystore.csv')
```

```
# Handle missing values in the Rating column by replacing NaNs with the mean rating
```

```
imputer = SimpleImputer(missing_values=np.nan, strategy='mean')
```

```
df.iloc[:, 2:3] = imputer.fit_transform(df.iloc[:, 2:3])
```

```
# Drop remaining NaN values
```

```
df = df.dropna()
```

```
df.head()
```

Output:

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
0	Photo Editor & Candy Camera & Grid & ScrapBook	ART_AND_DESIGN	4.1	159	19M	10,000+	Free	0	Everyone	Art & Design	January 7, 2018	1.0.0	4.0.3 and up
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Understanding Conditions in Pandas

Instead of iterating through each row, pandas allows us to apply conditions directly on columns.

Selecting a Column

We can extract a column as follows:

```
df['App']
```

```
df['Category']
```

Applying a Condition

For example, to check which rows belong to the **Art & Design** category:

```
df['Category'] == 'ART_AND_DESIGN'
```

This returns a **boolean series**, where:

- True means the app belongs to *Art & Design*.
- False means it does not.

Filtering Rows Based on a Condition

To filter all rows where the category is **Art & Design**:

```
df[df['Category'] == 'ART_AND_DESIGN']
```

This extracts all apps in the *Art & Design* category.

Q1. How many free apps are there in ART_AND_DESIGN?

We filter the dataset using conditions instead of loops:

```
df_pr = df[df['Category'] == 'ART_AND_DESIGN']
```

```
print(len(df_pr[df_pr['Type'] == 'Free']))
```

Output:

Q2. How many apps are there in ART_AND_DESIGN with rating more than 4.5?

We apply a condition to check ratings > 4.5:

```
df_pr = df[df['Category'] == 'ART_AND_DESIGN']
print(len(df_pr[df_pr['Rating'] > 4.5]))
```

Output:

22

Q3. How many apps are there in FAMILY with rating more than 4.5 and Free?

Now, we apply **three conditions** simultaneously:

1. Category = FAMILY
2. Rating > 4.5
3. Type = Free

```
df_pr = df[df['Category'] == 'FAMILY']
df_pr = df_pr[df_pr['Rating'] > 4.5]
```

```
len(df_pr[df_pr['Type'] == 'Free'])
```

Output:

314

Q4. List all the free apps with rating more than 4.5 and category is FAMILY?

Instead of counting, let's list all matching apps.

```
df_pr = df[df['Category'] == 'FAMILY']
df_pr = df_pr[df_pr['Rating'] > 4.5]
```

```
result = df_pr[df_pr['Type'] == 'Free'].values
```

```
for i in result:
```

```
    print(i[0])
```

This returns a **filtered DataFrame** with all apps that match the condition.

Output:

These are the free apps where Category is FAMILY with rating more than 4.5.

Super ABC! Learning games for kids! Preschool apps

Candy Pop Story

Dog Run - Pet Dog Simulator

Puzzle Kids - Animals Shapes and Jigsaw Puzzles

No. Color - Color by Number, Number Coloring

Garden Fruit Legend

Piano Kids - Music & Songs

Duolingo: Learn Languages Free

Super ABC! Learning games for kids! Preschool apps

Drawing for Kids Learning Games for Toddlers age 3

Baby ABC in box! Kids alphabet games for toddlers!

Henry Danger Crime Warp

Fruit Cube Blast

Toddler Kids Puzzles PUZZINGO

Fuzzy Seasons: Animal Forest

Dog Run - Pet Dog Simulator

My Oasis - Calming and Relaxing Idle Clicker Game

Ninja Dash - Ronin Jump RPG

Monster High™ Minis Mania

Monica Toy TV

Sworkit Kids - Fitness Meets Fun

Crazy Colors: Bubbles Matching

Pino chess

Duolingo: Learn Languages Free

Animal Jam - Play Wild!

Dog Sim Online: Raise a Family

Color by Number - Draw Sandbox Pixel Art

Build a Bridge!

Find a Way: Addictive Puzzle

CompTIA Exam Training

Gymnastics Superstar - Spin your way to gold!

Hungry Hearts Diner: A Tale of Star-Crossed Souls

Dragon Sim Online: Be A Dragon

Cat Sim Online: Play with Cats

Clash of Clans

Clash Royale

Learn C++

C Programming Zone

All C Programs

codeEazy - C/C++/JAVA
C Functions
Learn Basic C - For Beginners
C Prowess
101 C Programming Problems
C BOOK-C PROGRAMS
Eternium
Harris J Lyrics
2048 BTS J Hope KPop Puzzle Game
J. Cole Albums (2007-2017)
K-Otic Universe
CASE K-12 Math English Science
K-Spapp, the K-Space app
EXO-L
Fantastic Chefs: Match 'n Cook
Twitch: Livestream Multiplayer Games & Esports
Channel U
Get Free V-bucks_fortnite Hints
Easy V-Bux free
X-Wing Squadron Builder
JW Caleb y Sofia
PewDiePie's Tuber Simulator
Z-Empire: Dead Strike
Angry Birds Blast
Angry Birds Match
Angry Birds 2
Alchemy Classic Ad Free
Nasihah - Chalo ae Pyaara
9GAG: Funny Gifs, Pics, Memes & Videos for IGTV
Orbita AI — Exciting mobile puzzles & riddles
Learn Artificial Intelligence
Animal Jam - Play Wild!
AJ Official
Little Pets Animal Guardians
AJ Rafael Music Lessons
Weapon stripping 3D
Al Jazeera TV
Al Quran Audio (Full 30 Juz)
Quran for All (Al-Huda Int.)
Ayat - Al Quran
Quran Juz-30 - Mahad al Zahra
Al Quran MP3 - Quran Reading®
Al-Quran Ahmad Saud Offline
Virtual lover
virtual lover 3D

Bubbu – My Virtual Pet
AP® Guide
AP Mobile
Study AP World History
AP Chemistry Prep: Practice Tests and Flashcards
AP World History: Practice Tests and Flashcards
AP Biology Prep: Practice Tests, Flashcards
AP European History: Practice Tests and Flashcards
Adventure Quest World Mobile Quiz
AQ Ria Retail
Accounting Quiz (AQ) Malaysia
Wowkwis aq Ka'qaquj
Questland: Turn Based RPG
Grow Stone Online : 2d pixel RPG, MMORPG game
Egg, Inc.
Choices: Stories You Play
Aw Restaurant
AW Training
Anime Expo 2018
INTERACTIVE CALCULUS FOR MATHS AND PHYSICS
Pinoy Kid Song Ako ay may Lobo
Quran Khmer Offline AY
Permit Test AZ Arizona MVD DOT
Catalyst AZ
Az Video Status
Goody.ba
Learn alif ba ta
Arabic Alphabet Alif Ba Ta Wooden Blocks
Let's Learn Alif Ba Ta
Sufara.ba
Elif Ba - Learn The Holy Quran
Elifba Quran Learning Game
Railroad Radio Vancouver BC
BD Online Passport Application
BD Hospital's
Exam Result BD
BG Future School
Learn Bulgarian Free
Bh Public School
BH Challenge - Museum of the Jewish People
Bitmoji – Your Personal Emoji
Business Intelligence & Data
TED
English Grammar Test
HON. B.J. ACS COLLEGE ALE

BK Video Status
Audio Murli Glossary For BK
Baba Yaad Hai?(BK's)
BK murli today (Mobile Murli) - Gyan
BK Murli Dictionary (H to E)
BkEmu - BK-0010/11M emulator
Cerulean Heart
あなカレ【BL】無料ゲーム
BM SPM Practice
Township
Blood Pressure Diary
RoboPad
Br Shafi
Companion for Fortnite & Fortnite Battle Royale
Vlogger Go Viral - Tuber Game
Loved by King Bs
BS CS IT & SE
Marked by King Bs
COMSATS BOOK STORE FOR BS(CS)
Beauty Rental Shop
Sleeping Delivery
Wifi BT Scanner
BU Study
Barisal University App-BU Face
BU Students' Rep. Council
BU Calculator
BV Sridhara Maharaj
Quadratic Equation Solver with Steps and Graphs
Sandbox - Color by Number Coloring Pages
Sandbox Art-Sandbox Color by Number Coloring Pages
Color By Number - Sandbox Pixel Coloring Book
PixelDot - Color by Number Sandbox Pixel Art
PixPanda - Color by Number Pixel Art Coloring Book
Pixyfy: coloring by number coloring book
Draw Color by Number - Sandbox Pixel Art
Butterfly Pixel Art - coloring by number
Color by Number: Pixel Art
UNICORN - Color By Number & Pixel Art Coloring
FunnyPixels - Color by number Sandbox
No.Diamond – Colors by Number
Park Limousine: Realistic Limo Parking Simulator
CB News
CB Register
CD CHOICE TUBE
Yazdani Cd Center EllahAbad Official App

MCQ CE IT

Magana Jari ce Littafi Na Uku : Part 3 of 3

CF Life

All Info about Cg

CG FM

CG - Chemistry free

CG Jobs

BF CG Abstract Pictures

CI Dictionary

Sir C J New Primary School

Louis CK

Tic Tac CK

Castle Clash: Heroes of the Empire US

Cartoon Network Match Land

Super Slime Blitz - Gumball

Surely You Quest - Magiswords

We Bare Bears Match3 Repairs

Champions and Challengers - Adventure Time

CN Resident

CP Channel

Foothills CP

Cp Calculator for Evolution:GO

ClanManagerTT2

Christian Questions Podcast

CR & CoC Private Server - Clash Barbarians PRO

Ultimate Clash Royale Tracker

Servidor Privado de CR y CoC - Light Royale

Deck Shop for Clash Royale

Cr Aviation Academy

CS Interview Questions (TechQ)

DU CS Lectures - Learn to Code for Free

CS & IT Interview Questions

CA CS Network

CT Scan Generations

CT Brain Interpretation

Permit Test Connecticut CT DMV

Cy-Reader

MadLipz

DB Tools

DB for Hustle Castle

MARVEL Future Fight

DC N COMPANY ENTERTAINMENT RADIO!

Driver Permit Test Prep DC DMV Driver's License Ed

Coloring Book for Me & Mandala

Anime Love Story Games: ✨ Shadowtime ✨

DH-UFO
Idle Heroes
Duolingo: Learn Languages Free
PK and DK Audio App
dk
Account Class-12 Solutions (D K Goel) Vol-2
DL Hughley
DM Buddy » Learn Digital Marketing
Disciple Maker's (DM) Lab
DN Employee
Dragon Nest M
CritDice - Dice Roller
Maher EL Mouaikly - Offline
Darkness Rises
Latest DP and Status Pro 2018
Faceoff DP Battle App
RPG Dragon Lapis
DQSalmaan - A fan made App
DS
MegaN64 (N64 Emulator)
DS-Students
DT Driving Test Theory
WPBS-DT
Du Chinese – Mandarin Lessons
DV ASSIST
DV Lottery Simulator
DV-2019 UK/British
DW Audio
German Listening
Piczle Lines DX
DX Simulation for OOO Dx Belt
DX Simulation for Kabuto Henshin Belt 2018
DX Simulation for Double Dx Henshin Belt 2018
KPS CHAUHAN DY/DX
DYP SOET
Typical EA Game
EC Calgary
EC Fairgrounds
GATE Syllabus for EC 2018 & Notifications
English Conversation Courses
Dance Magic Fashion Style Games
EI国际
TeamWard – live help for LoL
Master E.K
Ek Onkar

Pyaar Ek Dhoka
Ek Bander Ne Kholi Dukan
Toilet Ek Prem Katha Songs Lyrics in Hindi
Sanu Ek Pal Chain Song Videos - RAID Movie Songs
Ek Qissa He Quran Se (Qurani Waqiyat)
Lyrics of Ek Paheli Leela
Exiled Kingdoms RPG
Shabad Gurubani Punjabi mp3 free - Ek Onkar Satnam
El Fali
OWLIE BOO
Narcos: Cartel Wars
Hello English: Learn English
Learn Top 300 English Words
Masha and the Bear. Games for kids
Masha and The Bear Jam Day Match 3 games for kids
Masha and the Bear: Good Night!
EP Radio
Super Doctor 2 - My Newborn Baby ER Injection
Open Heart Surgery Hospital ER: Crazy Doctor Sim
YEBIS OpenGL ES 3.0 Tech Demo
Advanced calculator fx 991 es plus & 991 ms plus
Castle Clash: Epic Empire ES
School scientific calculator fx 500 es plus 500 ms
Lettre et demande d'emploi
daskal.eu
PokeType - Dex
Pink Guy - Ey B0ss
Ey Sey Storytime រឿងនិទានកាតស៊ី
Fallout Shelter
Story Time FD
FD VR Music Videos - MTV Pop and Rap in 360
Fire Emblem Heroes
Safe Santa Fe
POLARIS 2MFG
Puzzles Volvo FH 16 Trucks
Jigsaw Volvo FH 16 Trucks
FH School
FK Željezničar Izzy
Permit Test FL Florida DHSMV
FL Lottery Results
Scratch-Off Guide for FL Lotto
Canvas FL
Florida Wildflowers
How it Works: FN SCAR assault rifle
Ninja FN Button

In data analysis, we often need to group similar data and perform operations such as **counting, averaging, or summing values** within each group. The **groupby()** function in Pandas allows us to do this efficiently.

- **Splitting** : It is a process in which we split data into group by applying some conditions on datasets.
- **Applying** : It is a process in which we apply a function to each group independently
- **Combining** : It is a process in which we combine different datasets after applying groupby and results into a data structure

The following image will help in understanding a process involve in Groupby concept.

1. Group the unique values from the Team column

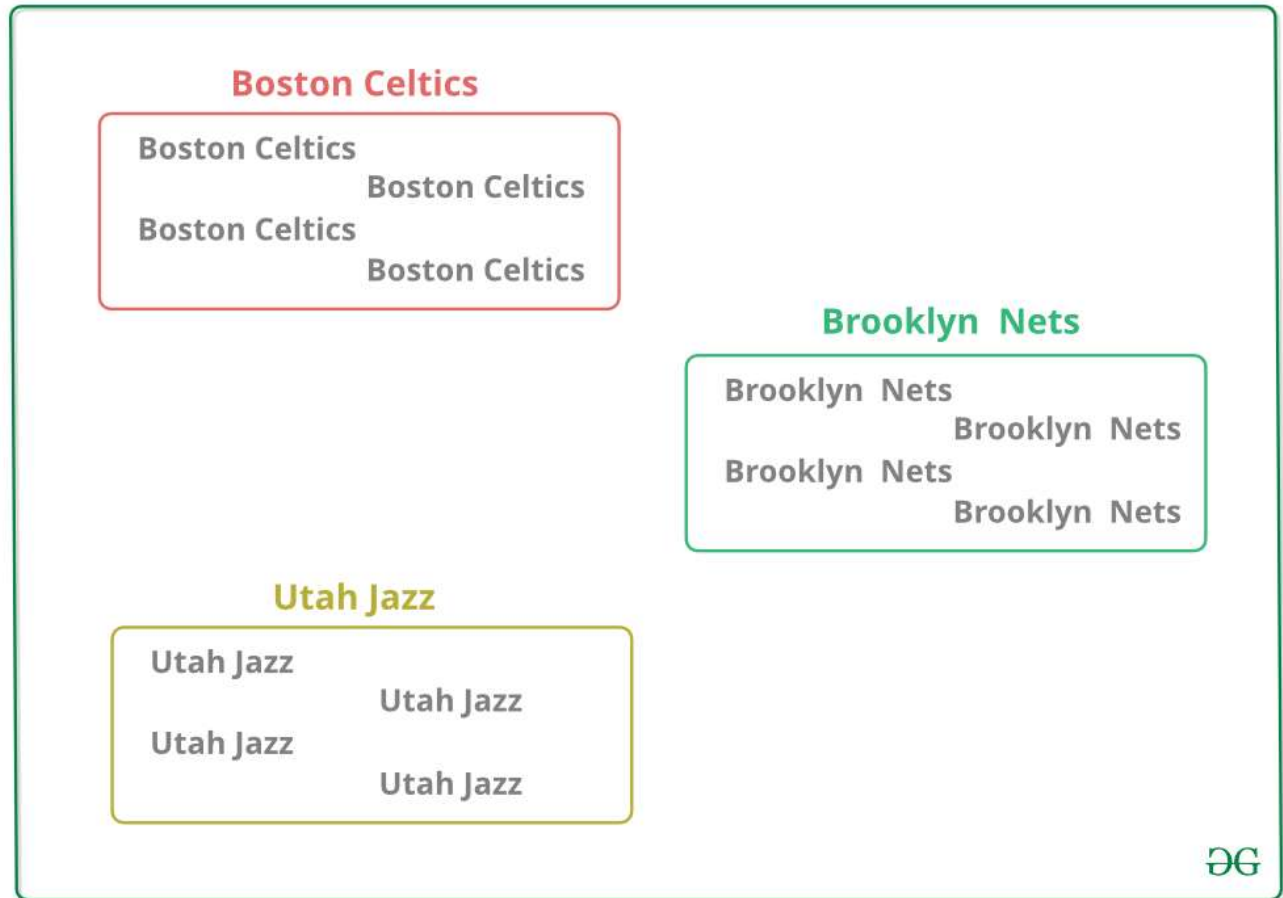
	Name	Team	Position	Age	Weight
0	Avery Bradley	Boston Celtics	PG	25.0	180.0
1	Jae Crowder	Boston Celtics	SF	25.0	235.0
2	John Holland	Boston Celtics	SG	27.0	205.0
3	R.j. Hunter	Boston Celtics	SG	22.0	185.0
4	Sergey Karasev	Brooklyn Nets	SG	22.0	208.0
5	sean Kilpatrick	Brooklyn Nets	SG	26.0	219.0
6	Shane Larkin	Brooklyn Nets	PG	23.0	175.0
7	Brook Lopez	Brooklyn Nets	C	28.0	275.0
8	Chris Johnson	Utah Jazz	SF	26.0	206.0
9	Trey Lyles	Utah Jazz	PF	20.0	234.0
10	Shelvin Mack	Utah Jazz	PG	26.0	203.0
11	Raul Pleiss	Utah Jazz	PG	24.0	179.0

Boston Celtics
Boston Celtics
Boston Celtics
Boston Celtics

Brooklyn Nets
Brooklyn Nets
Brooklyn Nets
Brooklyn Nets

Utah Jazz
Utah Jazz
Utah Jazz
Utah Jazz

2. Now there's a bucket for each group



3. Toss the other data into the buckets

	Name	Team	Position	Age	Weight
0	Avery Bradley	Boston Celtics	PG	25.0	180.0
1	Jae Crowder	Boston Celtics	SF	25.0	235.0
2	John Holland	Boston Celtics	SG	27.0	205.0
3	R.j. Hunter	Boston Celtics	SG	22.0	185.0
4	Sergey Karasev	Brooklyn Nets	SG	22.0	208.0
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6	Shane Larkin	Brooklyn Nets	PG	23.0	175.0
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8	Chris Johnson	Utah Jazz	SF	26.0	206.0
9	Trey Lyles	Utah Jazz	PF	20.0	234.0
10	Shelvin Mack	Utah Jazz	PG	26.0	203.0
11	Raul Pleiss	Utah Jazz	PG	24.0	179.0

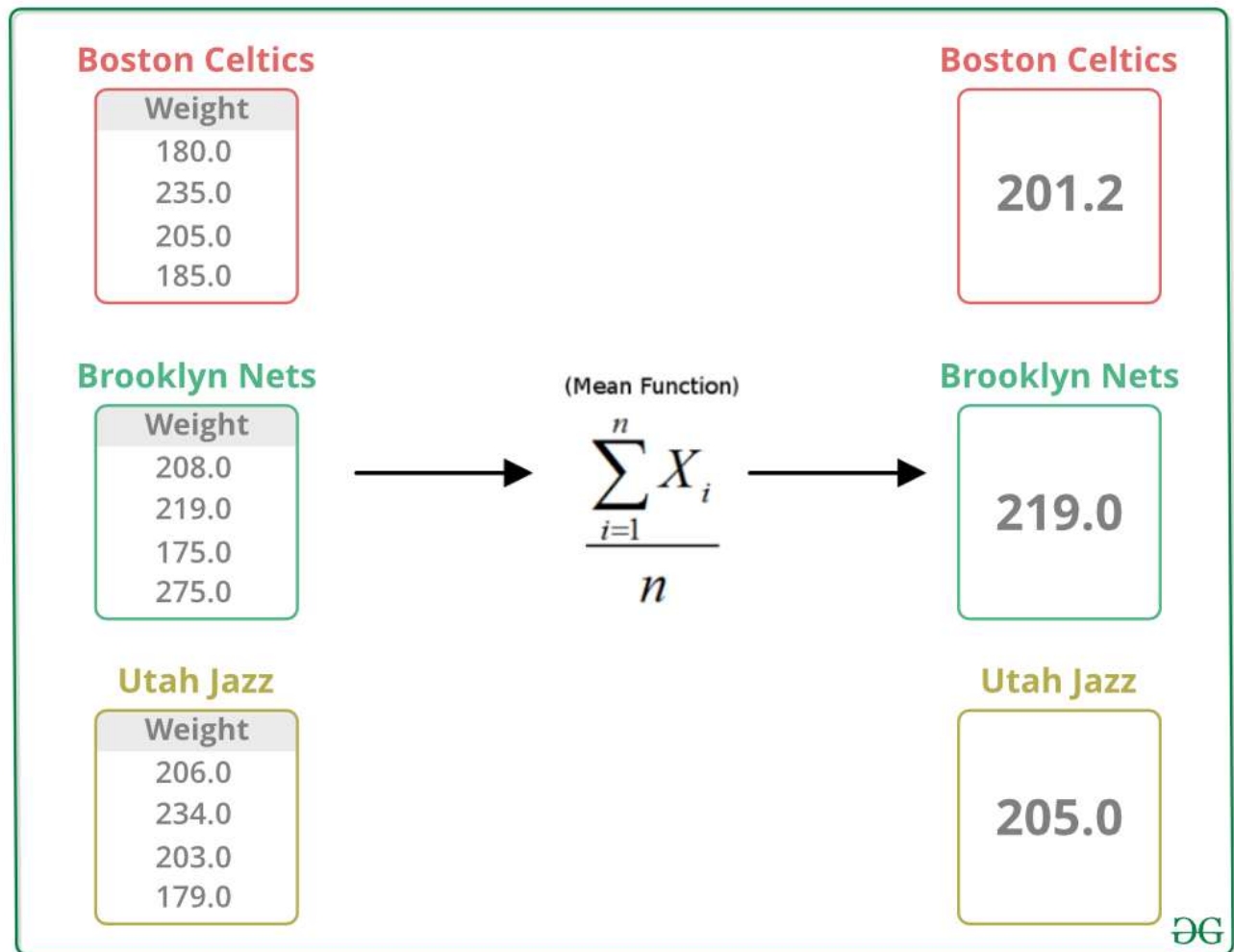
Name	Position	Age	Weight
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Brook Lopez	C	28.0	275.0

Name	Position	Age	Weight
Chris Johnson	SF	26.0	206.0
Trey Lyles	PF	20.0	234.0
Shelvin Mack	PG	26.0	203.0
Raul Pleiss	PG	24.0	179.0

OG

4. Apply a function on the weight column of each bucket.



Now let us see the implementation of this in our **googleplaystore** dataset.

Dataset Preparation

Before we apply grouping and sorting operations, we first **load and clean the dataset**.

```
import numpy as np
```

```
import pandas as pd
```

```
from sklearn.impute import SimpleImputer
```

```
df = pd.read_csv('googleplaystore.csv')
```

```
# Handle missing values in the Rating column by replacing NaNs with the mean rating
```

```
imputer = SimpleImputer(missing_values=np.nan, strategy='mean')
```

```
df.iloc[:, 2:3] = imputer.fit_transform(df.iloc[:, 2:3])
```

Drop remaining NaN values

df = df.dropna()

df.head()

Output:

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
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4	Pixel Draw - Number Art Coloring Book	ART_AND_DESIGN	4.3	967	2.8M	100,000+	Free	0	Everyone	Art & Design;Creativity	June 20, 2018	1.1	4.4 and up

Let us handle the null values again before getting started.

Dataset Columns

The dataset contains:

- App: Name of the application.
- Category: The category of the app (e.g., ART_AND_DESIGN, FAMILY).
- Rating: The user rating.
- Reviews: Number of reviews received.
- Size: App size.
- Installs: Number of installs.
- Type: Free or Paid.
- Price: Cost of the app.
- Content Rating: Age group suitability.
- Genres, Last Updated, Current Ver, Android Ver.

Sorting Values in Pandas

Before diving into groupby(), let's understand **sorting** in Pandas.

Q1. Name the apps that are in ART_AND_DESIGN with rating more than 4.5 in descending order WRT Ratings?

We filter apps based on category and rating and **sort them** in ascending order.

```
df_pr = df[df['Category'] == 'ART_AND_DESIGN']
```

```
df_pr = df_pr[df_pr['Rating'] > 4.5]
```

```
df_pr.sort_values(by='Rating', ascending=True).head()
```

Output

This will return the **top 5 apps in ascending order** of their ratings.

	App	Category	Rating	Reviews	Size	Installs	Type	Price	Content Rating	Genres	Last Updated	Current Ver	Android Ver
13	Mandala Coloring Book	ART_AND_DESIGN	4.6	4326	21M	100,000+	Free	0	Everyone	Art & Design	June 26, 2018	1.0.4	4.4 and up
19	ibis Paint X	ART_AND_DESIGN	4.6	224399	31M	10,000,000+	Free	0	Everyone	Art & Design	July 30, 2018	5.5.4	4.1 and up
46	Install images with music to make video without...	ART_AND_DESIGN	4.6	1070	26M	100,000+	Free	0	Everyone	Art & Design	November 14, 2017	1.6	4.1 and up
2	U Launcher Lite – FREE Live Cool Themes, Hide ...	ART_AND_DESIGN	4.7	87510	8.7M	5,000,000+	Free	0	Everyone	Art & Design	August 1, 2018	1.2.4	4.0.3 and up
4764	X Launcher Prime: With OS Style Theme & No Ads	ART_AND_DESIGN	4.7	149	3.5M	1,000+	Paid	\$1.99	Everyone	Art & Design	July 30, 2018	1.2.2	4.1 and up

Q2. Which Category Has the Highest Average Rating?

To find the category with the **highest average rating**, we group by Category and compute the **mean rating**.

```
df.groupby('Category').mean()['Rating'].sort_values(ascending=False)
```

Output:

```
Category
EDUCATION      4.387778
ART_AND_DESIGN  4.368438
EVENTS         4.363647
BOOKS_AND_REFERENCE  4.311537
PERSONALIZATION  4.306873
GAME           4.282506
```

PARENTING	4.282223
HEALTH_AND_FITNESS	4.266296
BEAUTY	4.260882
SHOPPING	4.254052
SOCIAL	4.248001
WEATHER	4.239675
SPORTS	4.218404
PRODUCTIVITY	4.208287
HOUSE_AND_HOME	4.196819
FAMILY	4.192490
PHOTOGRAPHY	4.192179
AUTO_AND_VEHICLES	4.190824
MEDICAL	4.190167
LIBRARIES_AND_DEMO	4.182938
FOOD_AND_DRINK	4.170709
COMMUNICATION	4.163842
COMICS	4.156445
BUSINESS	4.145987
NEWS_AND_MAGAZINES	4.142993
FINANCE	4.139108
ENTERTAINMENT	4.126174
TRAVEL_AND_LOCAL	4.119716
LIFESTYLE	4.112427
VIDEO_PLAYERS	4.074858
TOOLS	4.065970
MAPS_AND_NAVIGATION	4.065061
DATING	4.007864

Name: Rating, dtype: float64

EDUCATION has the highest average rating (4.38).

DATING has the lowest average rating (4.00).

Using groupby() in Pandas

The groupby() function is **powerful for aggregating data** based on a common column. It helps answer questions such as:

- Which category has the highest average rating?
- How many apps exist in each category?
- How many paid/free apps are there per category?

Q3. How many paid apps are there in each category in Descending Order?

To count the **number of paid apps per category**, we:

1. Filter apps where Type = **Paid**.
2. Use groupby() on the Category column.

3. Count the number of occurrences.

```
df_pr = df[df['Type'] == 'Paid']
```

```
df_pr.groupby('Category').count()['Type'].sort_values(ascending=False)
```

Output:

```
Category
FAMILY      190
MEDICAL     109
GAME        83
PERSONALIZATION  82
TOOLS       77
BOOKS_AND_REFERENCE  28
PRODUCTIVITY  28
COMMUNICATION  27
SPORTS      24
PHOTOGRAPHY  22
LIFESTYLE   19
FINANCE     17
HEALTH_AND_FITNESS  16
BUSINESS    14
TRAVEL_AND_LOCAL  12
WEATHER     8
DATING      7
MAPS_AND_NAVIGATION  5
EDUCATION   4
VIDEO_PLAYERS  4
AUTO_AND_VEHICLES  3
SOCIAL      3
ART_AND_DESIGN  3
NEWS_AND_MAGAZINES  2
PARENTING    2
FOOD_AND_DRINK  2
SHOPPING     2
ENTERTAINMENT  2
LIBRARIES_AND_DEMO  1
EVENTS       1
```

Name: Type, dtype: int64

- The **FAMILY** category has the most **paid** apps (**190**).
- The **EVENTS** category has the least **paid** apps (**1**).

Q4: How Many Free Apps Are There in Each Category?

To count **free apps** instead of paid ones:

```
df_pr = df[df['Type'] == 'Free']
```

```
df_pr.groupby('Category').count()['Type'].sort_values(ascending=False)
```

This will return a similar list, but for **free apps** instead.